Is Perceptual Diminishing Returns Organized by the Cones? A Cross-Validated Alignment Test on Public Colour-Difference Data

Anonymous Authors

follow-up to Bujack et al., PNAS 119:e2119753119 (2022)

Abstract

Perceived colour differences show *diminishing returns* (DR): along the neutral axis, small steps sum to more than the large step they span, so colour space is not Riemannian [2]. DR also appears on chromatic lines with location-dependent strength [25], which leaves open whether the nonlinearity is organised by the cone channels. We test this with public data only (91,896 gray-axis 2AFC triads, 12,000 MLDS triads, and Helm's and Ekman's hue-circle data), placed in a von Kries-normalised CIE 2006 cone space. A gauge-fixed nested family (additive M0 $\subset$ isotropic DR M1 $\subset$ per-cone DR M2) is compared out of sample, with per-dataset observation models and 10-fold cross-validation, against the best of six matched non-cone bases (opponent, sRGB, CIE RGB, CIELAB, CIE LCh), using selection-aware bootstrap inference. The 2022 gray-axis result replicates on held-out data (+418.2 log-likelihood units [234.3, 600.5]), and one global DR scaling generalises across datasets (+521.9 [322.0, 728.8]). Beyond it, no channel organisation is supported. Per-cone DR does not beat isotropic DR (-11.1 [-184.6, 102.8]), and the cone basis is never significantly better than the best control (-92.5 [-406.5, 61.8] against CIE LCh; significantly worse in five of eight sensitivity specifications). Synthetic recovery shows that per-cone curvature is not identifiable at the sampled directions, and the real held-out comparisons are several times more variable than any fitted model implies. Without a hue-curvature control, the data would have suggested a spurious S-specific compression. Public data therefore support an earned, scoped null: DR is global at this resolution, but its site (on the cones or after their combination) is not identifiable.

1 Introduction

For a century, perceived colour difference has been modelled as geodesic length in a Riemannian manifold [30], which implies additivity along geodesics: $D(A,B) + D(B,C) = D(A,C)$ for A, B, C in order. *Diminishing returns* (DR) is the strict subadditivity $D(A,B) + D(B,C) > D(A,C)$. Bujack et al. [2] demonstrated it on the neutral axis, so colour space is not Riemannian. Stark, Turton and Bujack [25] found DR on two chromatic lines and showed that its strength is *not constant*: one global scaling fitted on one line fails on the other. They then posed a question that two geodesics cannot settle: *is the nonlinearity organised by the cone channels?*

Why alignment matters. (i) *Compact prediction*: if DR decomposes into three per-cone functions plus one outer scaling, a handful of measurements generates DR everywhere, instead of an atlas measured direction by direction. (ii) *Mechanism*: cone alignment (L, M, S) would place a compression on each receptor signal *before* the signals are combined. Alignment with post-receptoral quantities

($L{+}M$, $L{-}M$, $S{-}(L{+}M)$) would place it *after* aggregation, at a different site. (iii) *Falsifiability*: if cone-aligned per-axis DR is no better than the same structure on standard non-cone axis systems, DR is not specifically cone-organised. So the control must be a *panel* of standard bases, and the comparison must be against its best member with selection-aware uncertainty. We calibrate these motivations to what the data support (§5).

What we do. With no new data, we place public human data in a stated, von Kries-normalised CIE 2006 cone space with a fixed receptor compression: 91,896 gray-axis 2AFC triads [2], 12,000 MLDS lightness triads [29], Helm's hue-circle distance estimates [14] and Ekman's spectral similarities [12]. We compare out of sample a gauge-fixed nested family (additive M0 $\subset$ isotropic DR M1 $\subset$ per-cone DR M2) against a matched panel M3 of six fixed non-cone bases. The comparison uses per-dataset observation models, 10-fold unit-disjoint CV and a paired double bootstrap. Three analyses are logically prior: a held-out replication of the 2022 gray-axis result, a data-type-specific hue-curvature control, and an identifiability analysis asking which granularity public data can resolve.

Findings in brief. The anchor replicates (R1$-$R0 = +418.2 [234.3, 600.5]; amplitude 2.35 vs 2.34 noise SDs in 2022), and one global DR scaling generalises across all data (M1$-$M0 = +521.9 [322.0, 728.8]). Beyond it, no channel organisation is established. Per-cone DR does not beat isotropic DR (-11.1 [-184.6, 102.8]), the best control (CIE LCh) predicts at least as well as the cone model (selection-aware -92.5 [-406.5, 61.8]), and the coarse $L{+}M \mid S$ split fails (-42.1 [-132.7, 26.6]). Eight pre-registered sensitivity specifications agree, with one exception: a significant anisotropy in S4, which traces to that specification's change of the dark-gray scale. In five of the eight, a non-cone basis is significantly better. Without the curvature control, Helm's hue circle would have produced a spurious S-specific compression. The identifiability analysis shows the limits of the null: per-cone curvature is not recoverable from these data, and the three-cone split is not identifiable even in synthetic data generated with strongly distinct per-cone DR (the L-vs-M split is detected in 12% of data sets). Cone and opponent structure *are* discriminable under the models' own noise, but the real held-out comparisons are 2.9–8.4 times more variable than the fitted models imply, and the real outcome lies outside every generator's predictive range.

Contributions and positioning. (1) To our knowledge, the first test of *alignment* of perceptual DR, as opposed to its existence or non-constancy, run against a matched panel with selection-aware inference. (2) An identifiability account: per-cone curvature is not recoverable, and cone-versus-opponent discrimination, although powered under the models' own noise, is swamped by unmodelled variability in the real data. (3) A held-out replication of the 2022 result on two independent gray datasets. (4) Curvature controls matched to each data type. DR and its non-constancy are results of [2, 25], and mere anisotropy (M2>M1) would largely restate the latter. Our object is the cone-versus-control comparison, and its outcome here is an earned, scoped null.

2 Methodology

A normalised CIE 2006 cone space. Cone excitations use the CIE 2006 2° LMS fundamentals [6, 26] (energy units, peak-normalised; CVRL `linss2_10e_1` [8]). The stimuli are specified in CIE 1931 xyY, so we derive $\mathbf{M}$ ($\mathrm{LMS} = \mathbf{M}\,\mathrm{XYZ}$) by least squares over 390–780 nm. The L, M rows regress on $(\bar{x}, \bar{y}, \bar{z})$; the S row regresses on $\bar{z}$ only (Smith–Pokorny [24]). The constraint was adopted, before any fitting, because the pre-registered spectral check showed that the unconstrained fit assigns spurious S excitation to long-wavelength lights (Appendix A). Per-cone R^2 is 0.9992/0.9992/0.9769. Ekman's monochromatic lights take their exact excitations from the fundamentals at each wavelength. The unconstrained, everything-through-$\mathbf{M}$ pipeline exactly as pre-registered is reported as S8. Excitations are von Kries-normalised to each dataset's reference white, $c_i = \mathrm{LMS}_i/\mathrm{LMS}_{i,w}$, giving cone contrasts in which the white is $(1, 1, 1)$ and every neutral stimulus is $(Y/Y_w)(1, 1, 1)$. The whites are Illuminant C for Helm's chips, equal-energy E for Ekman's lights (D65 in S6) and D65 for the neutral data. Each signal then passes through the same fixed receptor compression $c_i' = 116\,h(c_i) - 16$, with h the CIE lightness function. On the neutral axis this gives $L' = M' = S' = L^*$, so the 2022 gray-axis analysis is an exact special case. Model inputs are cone-response differences $\Delta\mathbf{c} = (\Delta L', \Delta M', \Delta S')$. Because the model family is closed under per-axis rescaling and h is homogeneous above its toe, cone-model predictions do not depend on the relative $L{:}M{:}S$ scaling for

stimuli above the toe. The luminance-weighted variant (S4) is *not* such a rescaling for the darkest neutral stimuli, which it pushes into the toe.

Model family and gauge. For a basis with fixed axes $\mathbf{u}_k$, let $x_k = \langle \mathbf{u}_k, \Delta\mathbf{c} \rangle / s_k$, where s_k is a fixed design RMS computed from stimulus coordinates only. This is per-axis scaling, not a rotation. Every model has the form

$$D = F_a\Big(\sum_{k=1}^{3} w_k\, \psi(|x_k|; \kappa_k)\Big), \qquad \psi(u;\kappa) = \frac{\log(1+\kappa u)}{\log(1+\kappa)}, \qquad F_a(y) = \frac{\log(1+ay)}{a}, \tag{1}$$

with $\psi(u;0) = u$ and $F_0(y) = y$. *Gauge:* $F(0) = 0$ and $F'(0) = 1$. The outer function is restricted to the one-parameter concave log family of [2] ($a \geq 0$), so the flexible shape sits in the per-axis functions. Each inner function is pinned at the reference difference $u = 1$ ($f_k(1) = w_k$), which makes κ_k a pure shape parameter. $\sum_k w_k = 1$, and the response scale is a per-dataset nuisance. Multiplying an axis by λ is absorbed exactly by $\kappa_k \mapsto \kappa_k/\lambda$ with a weight change, so axis normalisation cannot favour a basis. Curvature is summarised by the bounded index $\rho_k = \log(1+\kappa_k)/\kappa_k$ (1 linear, $\to 0$ strongly compressive). **M0** (additive): cone axes, $\kappa \equiv 0$, $a = 0$, a weighted city-block metric. **M1** (isotropic DR): $\kappa \equiv 0$, a free. **M2** (per-cone DR): $\kappa_L, \kappa_M, \kappa_S$ and a free (6 structural parameters). These are nested: M0⊂M1⊂M2. **M3** (matched non-cone panel): the M2 form, with 6 structural parameters, on fixed, disclosed bases. These are (i) DKL-style opponent [10] (luminance, $L-M$ and $S-$luminance, in von Kries units with the white's luminance weights); (i′) an equal-weight opponent variant (extra member); (ii) linear sRGB and (iii) CIE 1931 RGB channels, as fixed linear maps of the adapted cone signals; (iv) CIELAB on its own coordinates $(\Delta L^*, \Delta a^*, \Delta b^*)$; and (v) CIE LCh on $(\Delta L^*, \Delta C^*_{ab}, \Delta H^*_{ab})$ (Appendix A.1). Nothing is rotated or linearised from data, so M3 never nests M2. *Pre-registered secondary models:* per-basis additive and isotropic models $\mathrm{M0}_B$ and $\mathrm{M1}_B$, used for the difference-in-differences $(\mathrm{M2} - \mathrm{M1}) - (\mathrm{M3}_B - \mathrm{M1}_B)$, which separates alignment of the nonlinearity from alignment of the L1 aggregation; the granularity ladder G1 (common κ) and C2 ($\kappa_L{=}\kappa_M \neq \kappa_S$, the coarse "$L{+}M$ vs S" split); 18 matched two-channel controls (each non-cone basis with two curvatures tied); and an orientation supplement rotating the L, M axes by θ in the von Kries L–M plane.

Observation models and held-out likelihood. The structural parameters (w, κ, a) are shared across datasets, and each dataset has its own observation model and nuisance parameters. Response scales are never pooled into one likelihood. *Gray 2AFC:* $P(\text{lighter test more different}) = \Phi(\beta_g[D(r,t_2) - D(r,t_1)])$ [2, 28]. *MLDS:* $P((s_2,s_3) \text{ more different}) = \Phi(\beta_o[D(s_2,s_3) - D(s_1,s_2)])$ [21], with one β_o per observer. *Helm:* $\log \delta_{ps} = \log \alpha_s + \log D_p + \mathcal{N}(0, \sigma_s^2)$ per session. *Ekman:* $d_p = 1 - e^{-\beta_E D_p} + \mathcal{N}(0, \sigma_E^2)$ (Shepard's link [23]). The held-out log-likelihood ℓ sums $\log p(y \mid \hat{\theta}_{\text{train}})$ over datasets and held-out units: Bernoulli log-probabilities over trials, and Gaussian log-densities for Helm and Ekman. Model differences are unit-free and are always also reported per dataset. For pairwise data the outer F is aliased with the response link, so these data mainly constrain the inner, direction-specific structure.

Fitting, cross-validation and comparison. We use L-BFGS-B [5] with exact JAX gradients [1], bounds $\kappa \in [0, 10^4]$ and $a \in [0, 10^3]$, and 8 random starts plus a deterministic start. Bootstrap refits use a warm start, the deterministic start and 2 random starts. Folds are $K{=}10$ and unit-disjoint: triad-disjoint for gray 2AFC (stratified by reference L^*; no participant IDs exist) and MLDS, pair-disjoint for Helm and Ekman. Inside each fold we draw $B{=}100$ cluster-bootstrap resamples of the *training* units, paired across models by shared seeds, and refit every model on each. A delta's 95% CI comes from a paired double bootstrap ($R{=}4000$) that resamples both the training refit and the held-out units. Because the headline comparator is the panel's *best* member, the argmax is recomputed in every replicate. We report the CI of $\min_k(\mathrm{M2} - \mathrm{M3}_k)$, and treat C2 against the 18 two-channel controls the same way. Parameter CIs are percentiles over the 1000 training-fold refits. *Decision rules (pre-registered).* (a) Anisotropy: M2−M1 with CI above 0. (b) Cone alignment: selection-aware M2−best M3 with CI above 0 *and* M2-vs-opponent discrimination powered at ≥80% in synthetic data. (c) Granularity: the finest level of M1→G1→C2→M2 that improves on the next coarser level with CI above 0 and whose parameters are synthetically recoverable.

Identifiability and confound control. Identifiability is assessed four ways: collinearity of the cone-difference design (correlations, VIFs, direction spectrum), canonical correlations between the

cone and control feature sets, profiled observed information, and a *synthetic* study. In the synthetic study, data are simulated at the real stimuli, trial counts and observation models from six generators (fitted M2, a designed M2 with distinct per-cone curvature, M1, DKL opponent, best M3, C2; 100 data sets each). Each data set goes through the same CV and decision rules. Curvature is controlled by the method appropriate to each data type. For triads we keep only those within 10° of collinear in cone space. For the pairwise hue loci we compare arc-length inputs (before control: per-axis path length along the locus, treating it as a geodesic) with chord inputs (after control: the additive-metric curvature baseline). Where a locus has near-geodesic runs we also check DR on those pairs alone.

3 Experimental Setup

Datasets, sources and licences. Only the provided public data are used. (1) *Gray 2AFC* [2]: `lanl/color`, BSD-3 [3]. 91,896 crowdsourced trials over 320 triads (64 configurations $\times$ 5 reference levels L^*=30..70; differences 0–30), 249–304 responses per triad. (2) *MLDS* [29]: OSF `58jz3`, CC BY 4.0. Plain context only: 12,000 trials from 10 observers $\times$ 120 triads $\times$ 10 repeats, with check luminances (24.5–312.2 cd/m^2) from the matching files. (3) *Helm* [14]: distance estimates for all 45 pairs of 10 Munsell-value-6 chips (`smacof`, GPL-3 [9]; chromaticities from Helm's Table I, Illuminant C). We use the 11 colour-normal sessions and exclude the 5 colour-deficient ones. (4) *Ekman* [12]: mean similarity of 14 equal-brightness spectral lights, $d = 1 - s$ (91 pairs; `smacof`, GPL-3). Fundamentals and CMFs come from CVRL [8]. The gray sets sample only the achromatic direction; the chromatic sets sample the near-isoluminant plane at moderate (Helm) and spectral (Ekman) saturation.

Protocol. Held-out likelihood, folds (K=10, seed 20261004; `logs/folds.json`; each fold holds out about 32 gray triads, 12 MLDS triads, 4–5 Helm pairs and 9–10 Ekman pairs), training-fold bootstrap (B=100, so every model in the primary specification was fitted 1010 times) and selection-aware inference are as in §2. All models share folds, bootstrap indices and observation models. The pre-registered secondary specifications S1–S8 (Table 6) use full-training fits with test-unit bootstrap CIs. These ignore training variability: on P they are 0.59 times the width of the double-bootstrap CIs.

Gray-axis replication (anchor). Before any extension we refitted the 2022 analysis on the gray data alone, with the same folds and B=100. The models were linear L^* (R0, identical to M0 on the neutral axis), log-f on L^* (R1, identical to M1), a monotone Fritsch–Carlson [13] perceptual scale g with 4 control points as in 2022 (R0g), and g jointly with log-f (R1g). The MLDS analogues are W0 and W1 on the cube-root scale and W0g and W1g with a free scale. DR replicates on held-out data in every case: R1$-$R0 = +418.2 [234.3, 600.5], R1g$-$R0g = +829.2 [635.4, 1043.4], W1$-$W0 = +77.8 [43.3, 117.5] and W1g$-$W0g = +81.8 [44.3, 133.3]. The gray deltas are positive in 9/10 and 10/10 folds. In 2022-style within-triad splits, the 2022 ordering is reproduced (accuracy 0.886 $<$ 0.893 $<$ 0.908 $<$ 0.925 for R0, R1, R0g, R1g). On the 2022 normalisation the amplitude on g is $b = 2.35$ [2.22, 2.49] noise SDs (2022: 2.34). The log shape is $a = 3.27$ [2.09, 4.90] on L^* and 12.39 [8.83, 17.53] on g (2022: 5.34). The subadditivity ratio $f(30)/(2f(15))$ is 0.749 [0.717, 0.788] on L^* and 0.683 [0.664, 0.703] on g (2022 log fit 0.71; spline example 0.76). The replication is qualitatively exact, and its magnitude brackets the 2022 report (Appendix H, Fig. 5).

4 Experimental Results

Held-out comparison (Table 1, Fig. 1). One global DR scaling generalises: M1$-$M0 = +521.9 [322.0, 728.8]. The gain comes mostly from the gray 2AFC (+415.5) and MLDS (+77.9) data, with Helm adding +35.4 [$-$5.1, 79.9]; for Ekman ($-$6.8 [$-$15.4, 6.3]) the exponential link already saturates. Per-cone curvature does not improve on isotropic DR: M2$-$M1 = $-$11.1 [$-$184.6, 102.8], positive in 4/10 folds. The decomposition is informative. M2 helps Helm (+68.9 [$-$116.9, 159.7]) but significantly *hurts* the gray 2AFC data ($-$70.4 [$-$96.9, $-$13.6]). The per-cone curvatures that fit Helm's hue circle also act on the neutral axis, where $\Delta L'$=$\Delta M'$=$\Delta S'$, and mispredict gray-axis DR. Bases with a dedicated achromatic axis (opponent, CIELAB, LCh) avoid this cost; bases whose three axes all respond to gray (sRGB, CIE RGB) pay it too. *Cone alignment.* The cone model is not better than the best panel member: M2$-$best M3 = $-$92.5 [$-$406.5, 61.8] (selection-aware; best CIE LCh). The identity of the best control is itself uncertain. CIE LCh is the argmax in only 32% of replicates (CIE RGB 22%, CIELAB 15%), and a naive interval with CIE LCh fixed is [$-$356.3, 422.1]. The

Table 1: **Primary results** (specification P). Held-out log-likelihood ℓ is summed over 10 unit-disjoint test folds and four datasets; 95% CIs come from the paired double bootstrap over training-fold refits and held-out units (R=4000). $\Delta\ell > 0$ favours the first model. The cone-alignment row is selection-aware: the best control is re-selected in every replicate.

Quantity	Estimate	95% CI	Comparison	Notes
Gray-axis DR replication: R1−R0 held-out $\Delta\ell$	+418.2	[234.3, 600.5]	2022: concave f (a=5.34, b=2.34)	fitted g: R1g−R0g +829.2 [635.4, 1043.4]; a=12.4, b=2.35
Held-out fit, M0 (additive)	0.0	(reference)	$\Delta\ell$ vs M0	total held-out ℓ −65447.6
Held-out fit, M1 (isotropic DR)	+521.9	[322.0, 728.8]	$\Delta\ell$ vs M0	1000 bootstrap refits
Held-out fit, M2 (per-cone DR)	+510.8	[245.7, 723.4]	$\Delta\ell$ vs M0	1000 bootstrap refits
Held-out fit, M3 opponent (DKL)	+506.4	[201.2, 884.0]	$\Delta\ell$ vs M0	matched complexity (6 struct.)
Held-out fit, M3 opponent (equal-weight) (extra member)	+515.1	[206.1, 907.8]	$\Delta\ell$ vs M0	matched complexity (6 struct.)
Held-out fit, M3 sRGB	+539.2	[224.1, 864.8]	$\Delta\ell$ vs M0	matched complexity (6 struct.)
Held-out fit, M3 CIE RGB	+565.8	[270.2, 880.8]	$\Delta\ell$ vs M0	matched complexity (6 struct.)
Held-out fit, M3 CIELAB	+543.6	[242.9, 912.9]	$\Delta\ell$ vs M0	matched complexity (6 struct.)
Held-out fit, M3 CIE LCh	+603.3	[79.0, 892.8]	$\Delta\ell$ vs M0	matched complexity (6 struct.)
M2 vs M1 (anisotropy)	−11.1	[−184.6, 102.8]	sign −; n.s.	gray −70.4 [−96.9, −13.6], Helm +68.9
M2 vs best M3 (cone alignment, selection-aware)	−92.5	[−406.5, 61.8]	best: CIE LCh; n.s.	best in 32% of replicates; fixed-best CI [−356.3, 422.1]
Identifiability: coarse $L+M \mid S$ split C2 vs M1	−42.1	[−132.7, 26.6]	C2 vs best of 18 two-channel controls: −123.5 [−423.4, 3.5]	per-cone ρ CIs span [0, 1]; synthetic: L/M split found in 12%, M2 vs DKL powered (100%) but real CIs 2.9–8.4× wider
Hue-curvature control: Helm DR (M1−M0) after chord control	+35.4	[−5.1, 79.9]	before (arc): +64.0 [−39.2, 130.1]	triads 100% collinear; C2−M1: +109.7 (arc) → −42.1 (chord)

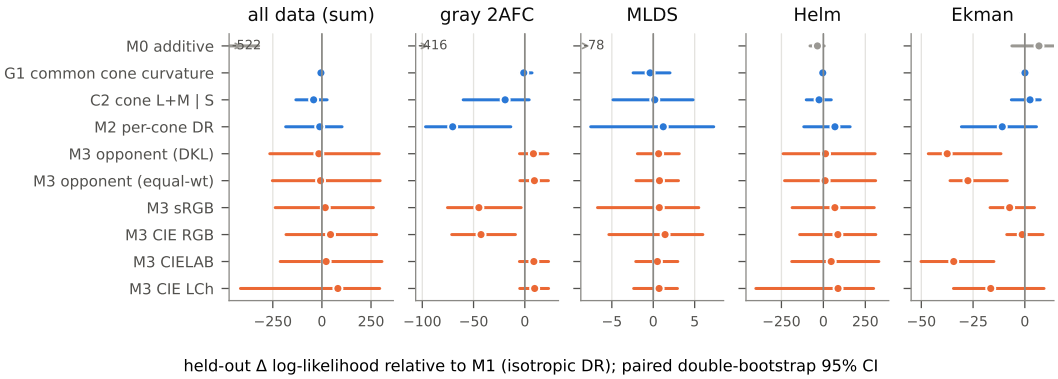


Figure 1: **Held-out comparison of M0, G1, C2, M2 and all six M3 panel members** relative to isotropic DR (M1), in total and per dataset (independent x-scales; 95% CIs). Cone family in blue, non-cone controls in orange, M0 in gray (off-scale values printed).

five required bases alone give −92.5 [−401.3, 65.2]. Pairwise, M2 is indistinguishable from every member, from M2−DKL = +4.4 [−317.0, 247.8] to M2−LCh = −92.5. The best of all 41 models is *isotropic* DR on LCh coordinates (Appendix C): bases differ in aggregation geometry, not in where the nonlinearity sits. Accordingly, no difference-in-differences CI excludes 0 (range −72.9 [−303.9, 32.4] to +20.1 [−203.1, 200.1]), and in no basis does per-axis curvature beat isotropic DR.

Fitted per-cone functions and distinctness (Fig. 2). M2 does not collapse to M1 at the point estimate: $\rho_L, \rho_M, \rho_S = 0.00, 0.85, 0.57$, with weights 0.052, 0.660, 0.288. But κ_L sits at its bound, a near-step of tiny weight, and no per-cone curvature is determined. The 95% intervals over 1000 refits are [0.00, 1.00], [0.00, 1.00] and [0.00, 1.00]. κ hits the upper bound in 12/18/2% of refits, and every pairwise ρ difference interval contains 0 ($\rho_L - \rho_M$: [−0.93, 0.89]). The distinctness check therefore fails: M2 is a genuine per-cone model, but the data do not determine distinct per-cone scalings. C2 is better behaved: $\rho_{L+M} = 0.95$ [0.72, 1.00] and $\rho_S = 0.53$ [0.00, 1.00].

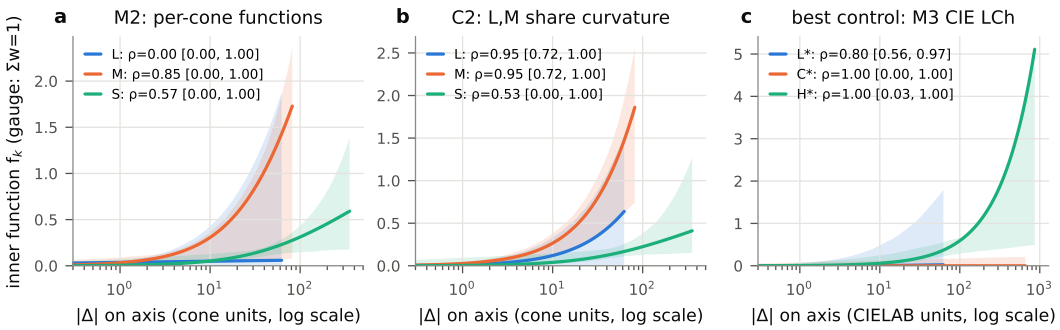


Figure 2: **Fitted inner functions** with 95% bands over 1000 training-fold refits, over the range of differences present in the data: (a) M2 per-cone f_L, f_M, f_S; (b) C2 (L, M share one curvature); (c) the best control (CIE LCh). Legends give the curvature index ρ with its CI.

Identifiability (Fig. 3). *Design.* The data span two subspaces. On the neutral axis $\Delta L' = \Delta M' = \Delta S'$ exactly, so the gray sets constrain only $f_L + f_M + f_S$ along one direction. The chromatic sets are near-isoluminant: signed $\Delta L'$ and $\Delta M'$ correlate -0.70 (Helm) and -0.08 (Ekman), and the $L-M$ direction carries only 2.5% and 2.4% of the design variance. Linear collinearity is moderate ($\mathrm{VIF}_{\max}$ 1.24, 1.59). The binding problem is that within the chromatic data the cone and DKL feature sets are nearly interchangeable (canonical correlations 1.000, 0.999, 0.423, against 0.974, 0.407, 0.022 for LCh). What separates them is mainly how the achromatic direction is represented, which only the gray data inform, through a cross-dataset tie. At the M2 optimum the profiled observed information gives log-scale SEs (0.50, 0.62) for κ_M, κ_S, far narrower than the bootstrap spread. *Orientation.* Rotating the L, M axes by θ (M2 at θ=0; the equal-weight $L \pm M$ pair at $45°$) changes held-out ℓ by at most 105.1 units, and none of the 17 rotations differs from the cone orientation (best θ=10°: +49.8 [−17.3, 122.5]; 45°: +31.9 [−102.3, 164.2]). The data cannot orient per-axis DR inside the $L \approx M$ subspace, which is exactly where cone and post-receptoral hypotheses differ. *Synthetic recovery and power* (**synthetic** data, 100 data sets per generator; Table 8). Under the designed distinct-DR generator ($\rho_L, \rho_M, \rho_S = 0.89, 0.40, 0.09$), ρ_S is recovered (median 0.10, 95% range 0.05–0.24) and ρ_M roughly (0.27–0.86), but ρ_L is not (0.01–1.00). The L-vs-M split (M2>C2) is detected in only 12% of data sets and anisotropy (M2>M1) in 50%. At its fitted size, the coarse split C2>M1 is detected in 2%. The isotropic generator yields false-positive anisotropy in 0%. Cone versus control *is* discriminable under the models' own noise. The cone model beats DKL in 100% of data sets generated by the fitted M2 (98% for the designed M2); DKL wins in 100% of its own data; and LCh data reject the cone model in 100%. The pre-registered 80% criterion is therefore met. Two caveats follow. First, an *isotropic* cone-metric truth already makes M2 beat DKL in 90% and the best panel member in 53% of data sets, so the panel test rewards cone *aggregation* even without channel-structured DR. Second, in a post hoc calibration the real held-out intervals are 2.9–8.4 times wider than the synthetic ones for M2−DKL (4.0–15.3 times for M2−best M3). At real-data precision, effects of the generators' size would go undetected: data from the fitted M2 exceed the real half-width in 0% of cases. The real M2−DKL value (+4.4) lies outside every generator's 95% range, between the cone-family generators (medians +28.8 to +81.2) and the opponent and LCh generators (−102.3, −86.7). The real data are thus far more variable than any fitted model implies.

Hue-curvature confound control (Fig. 4). *Triads:* every gray and MLDS triad is collinear in cone space (deviation $< 10^{-5}$ degrees), so the near-collinear restriction keeps 100% and 100% of triads and the gray-axis DR verdict is unchanged by construction. *Hue loci:* treating the locus as a geodesic inflates chromatic DR (Helm M1−M0: +64.0 [−39.2, 130.1] with arc inputs vs +35.4 [−5.1, 79.9] with chord inputs). More importantly, it manufactures channel structure. With arc inputs, C2−M1 = +109.7 [−92.8, 330.8] (Helm +152.1), with a near-maximal, tightly estimated S compression ($\rho_S = 0.00$ [0.00, 0.17]). With chord inputs it vanishes (C2−M1 = −42.1; ρ_S unidentified). Per-cone anisotropy behaves the same way (M2−M1: +85.1 before, −11.1 after), and cone alignment is null on both sides (before: −27.4 [−374.7, 158.8] vs opponent (DKL)). On a closed hue circle, DR along a geodesic arc and an additive metric on a curved locus are *aliased*; the flexible models even predict Helm better from arc inputs (Table 7). Our verdicts are therefore conditional on the pre-registered chord baseline. Where the locus is locally straight the aliasing can be broken: on Ekman's two near-geodesic runs (434–537 nm and 600–674 nm) outer DR adds nothing beyond the response link

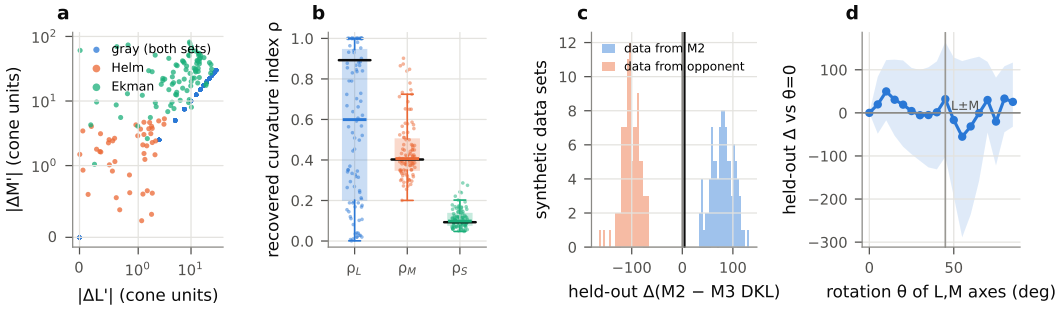


Figure 3: **Identifiability.** (a) Absolute cone differences of all pairs entering the likelihoods. Gray pairs lie on the diagonal; chromatic pairs mix L and M in near-fixed ratios. (b) *Synthetic* recovery of the per-cone curvature index under the designed distinct-DR generator (truth: black bars; 100 data sets). (c) *Synthetic* held-out $\Delta(\mathrm{M2} - \mathrm{M3}_{\mathrm{DKL}})$ for data generated by M2 vs by the DKL opponent model; the black line is the real-data value. (d) Real-data held-out ℓ of M2 with the L, M axes rotated by θ, relative to θ=0 (test-unit bootstrap 95% band).

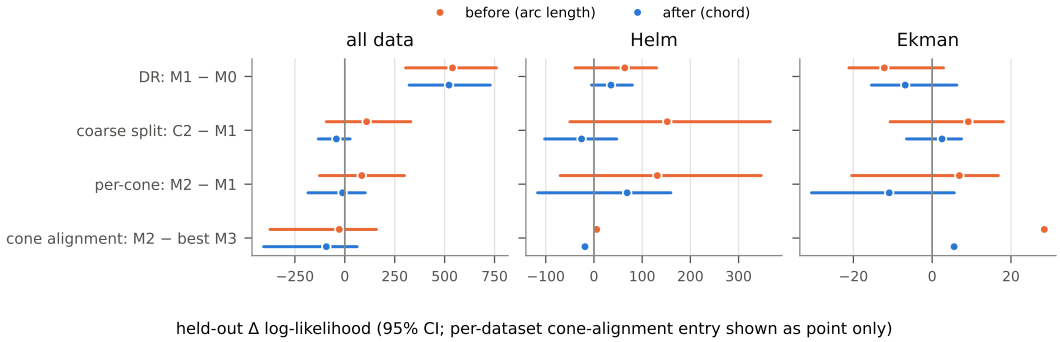


Figure 4: **Curvature control** before (arc inputs, orange) and after (chord inputs, blue), for DR, coarse and per-cone anisotropy and selection-aware cone alignment, in total and for the two hue-locus datasets. Without the control, apparent channel structure (C2) appears on Helm.

(M1−M0 = −1.30 [−3.49, −0.12] and −0.88 [−2.65, 0.00]). Helm's circle has 1 near-collinear triple, so no such check is possible there.

5 Analysis

Verdict (a): is DR channel-organised at all? Not demonstrably. Per-cone curvature does not improve on a single global scaling (M2−M1 = −11.1 [−184.6, 102.8]), nor do coarser splits (G1−M1 = −5.0 [−10.1, 4.6]; C2−M1 = −42.1 [−132.7, 26.6]), and in no basis does per-axis curvature beat isotropic DR. Of the eight secondary specifications (Table 6) only S4 gives a positive anisotropy CI (+150.9 [39.6, 282.4]). S4 is not a clean test: rescaling M by ≈0.3 moves dark neutral stimuli into the CIE toe, its gray terms swing by hundreds of units, and Helm alone drives its anisotropy. What the data *do* show is that DR strength differs between datasets. With dataset-specific outer scalings (S1), nothing is left for direction-dependent curvature (M2−M1 = +4.0 [−3.5, 11.8]). The subadditivity ratio $F(2y)/(2F(y))$ at each dataset's median pair is 0.62 for Helm, against 0.75 (gray), 0.75 (MLDS) and 0.89 (Ekman, whose link absorbs compression). This fits the non-constancy of [25], but here it is confounded with task and response scale, so we do not read it as direction dependence.

Verdict (b): is the structure cone-aligned? No. The cone model is never significantly better than the best member of the matched panel: −92.5 [−406.5, 61.8] in the primary analysis (best CIE LCh). In five of eight secondary specifications a non-cone basis is significantly better (S2, S3, S5, S7 and the confounded S4; test-unit CIs, which are optimistic). S2 removes every cross-dataset tie and still prefers sRGB (−73.6 [−178.6, −30.0]). The winning control is unstable (LCh, CIE RGB, sRGB or equal-weight opponent, depending on the specification); what is stable is that some standard non-cone basis matches or beats the cone basis. Two qualifications calibrate this. First, the bases differ mainly in aggregation geometry: the best model is isotropic DR on LCh coordinates, and

the difference-in-differences is null. "Not cone-aligned" therefore means "no evidence that the nonlinearity sits on the cone axes", not "it sits on LCh axes". Second, the null is not a failure to discriminate in principle. In model-based synthetic data the cone model beats the DKL opponent model in 100% (fitted M2) and 98% (designed M2) of data sets, and opponent and LCh truths are recognised in 100% and 100%, meeting the pre-registered 80% criterion. Two calibrations temper this. Real held-out intervals are 2.9–8.4 times wider than the synthetic ones (post hoc), and the real M2−DKL value (+4.4) lies outside every generator's 95% range. Moreover, an isotropic cone-metric truth already wins the panel test in 53% of synthetic data sets, so a positive panel verdict would not by itself have shown that the nonlinearity is cone-aligned. The difference-in-differences, which removes this aggregation advantage, is null throughout.

Verdict (c): which granularity is identifiable, and what it means for mechanism. Only the isotropic level. Global DR is resolved robustly (M1−M0 = +521.9 [322.0, 728.8]), with a CI excluding 0 in every specification, including the chromatic-only S3 (+148.8 [17.1, 284.4]). Neither the three-cone nor the coarse $L{+}M \mid S$ split is resolved. All per-cone curvature intervals span $[0, 1]$, the L–M orientation profile is flat, and in synthetic data with strongly distinct per-cone DR only ρ_S is recovered (0.10, 95% range 0.05–0.24), ρ_L is not (0.01–1.00), and the L-vs-M split is detected in 12% of data sets. The coarse split at its fitted size is detected in 2%. The empirical face of the pre- versus post-combination question is whether L and M carry distinct compressions or one compression of a combined signal. The data cannot answer it, because they vary L and M only in two fixed ratios: equal on the neutral axis and opposed at isoluminance. We therefore make *no* claim about the site of compression. The held-out cost on gray data of tying achromatic to chromatic DR (−70.4 [−96.9, −13.6]) vanishes once scalings are dataset-specific (S1), so it reflects cross-experiment pooling, not mechanism.

Failure analysis. (1) *Degenerate per-cone fits:* κ_L reaches its bound in the full fit and in 12% of refits, and w_L often collapses toward 0. (2) *Outer/inner trade-off:* on the neutral axis the outer and per-axis curvatures trade along a ridge (one replication fit hit a degenerate optimum, fixed by a deterministic start; Appendix B). (3) *Response-scale aliasing:* Ekman's link absorbs DR, and Helm's DR is aliased with ratio estimation. (4) *Curvature aliasing* on Helm's closed circle (an arc analysis manufactures S-specific compression). (5) *Wide intervals:* the cone-versus-control intervals are dominated by the 45 Helm pairs (M2−DKL: width 570 for Helm vs 85 for gray; 565 in total), where flexible per-axis fits mispredict individual held-out pairs, and real deltas are several times more variable than any fitted model implies. (6) Two implementation bugs (power-family offset; silent-axis scaling in per-dataset fits) affected only S5 and S2; both were fixed and rerun (Appendix G).

6 Related Work

Diminishing returns and the non-Riemannian program. Bujack et al. [2] showed DR on the neutral axis with a Thurstonian 2AFC triad analysis, a fitted perceptual scale g and a concave difference scaling f [27]. Because the neutral axis is believed to be a geodesic, this rules out any Riemannian metric. Stark, Turton and Bujack [25] repeated the design on two chromatic lines in CIELAB at $L^*{=}70$ (green–pink, blue–orange). They found DR on both lines and found that one global scaling does not transfer between them, and they proposed, but explicitly could not test, a per-cone decomposition. Bujack et al. [4] formalised hue, saturation and lightness in a non-Riemannian metric space and defined the neutral axis geometrically. Molchanov [22] quantified local inhomogeneity and anisotropy of CIELAB-based metrics within the Riemannian setting. *What we add:* not DR, its non-constancy or mere anisotropy, which belong to those works, but a test of *alignment*. We ask whether a per-axis decomposition of the nonlinearity predicts held-out judgments better in cone coordinates than in every standard non-cone system of equal complexity, and which granularity public data can identify.

Large colour differences and curvature. DR was named by Judd [16], and MacAdam fitted a power law to suprathreshold chromaticity differences [19]. Helm [14] found that his hue-circle distances needed logarithmic compression before MDS recovered a circle. Ekman's similarities [12] are the classic closed hue circle and an instance of Shepard's exponential law [23]; both data sets are in `smacof` [9]. Izmailov and Sokolov [15] contrasted arc length on a spherical colour manifold

with chord distance, which is the ambiguity our curvature control addresses. MLDS supplies suprathreshold interval scales [17, 21, 29].

Cone and post-receptoral coordinates. The CIE 2006 observer [6, 26], cone-contrast normalisation and the luminance, $L-M$ and $S-(L+M)$ mechanisms [10, 18, 20] are the natural candidates for a compressive stage before or after the cone signals combine. Modern uniform spaces compress each cone signal before an opponent transform (e.g. IPT [11]), whereas CIELAB [7] compresses X, Y, Z. Our panel uses exactly these fixed standard systems, so cone alignment is tested against the field's alternatives, not a straw man.

7 Conclusion

Findings. Using only public data in a documented, von Kries-normalised CIE 2006 cone space, we found the following. (1) The 2022 gray-axis result replicates out of sample on two independent gray datasets, with a matching amplitude (2.35 vs 2.34 noise SDs). (2) One global DR scaling generalises across all datasets and specifications. (3) Beyond it, no channel organisation is supported: per-cone, common and coarse $L+M \mid S$ curvature all fail to beat isotropic DR, and per-cone curvatures are not recoverable. (4) The cone basis is never significantly better than the best of six matched non-cone bases, and in several specifications a non-cone basis is significantly better, although which one varies. (5) Without a curvature control, Helm's hue circle would have suggested a spurious S-specific compression. The data thus support an *earned, scoped null*. DR is real and global at this resolution, but whether it sits on the cones, after their combination, or on perceptual-attribute axes cannot be resolved. Under the models' own noise cone and opponent structure would have been told apart, but real-data uncertainty exceeds the models' own by a factor of 2.9–8.4. The null therefore reflects two things: the stimulus design limits the per-cone split, and unmodelled heterogeneity within and across the pooled datasets limits the cone-versus-control contrast.

Contribution beyond prior work. The 2022 paper showed DR on one geodesic [2], "Now in Color" on two more together with non-constancy [25], and the CGF paper developed the geometry [4]. We add the first selection-aware test of *alignment*, an identifiability account of what public data can resolve (only the isotropic level), and evidence that apparent channel structure in hue-circle data is a curvature artefact. The motivation of §1 is scaled back accordingly: a compact per-cone generative model of DR is not supported by existing data, and no claim about the pre- versus post-combination site is warranted.

Limitations. The data span only the achromatic axis and the near-isoluminant plane. Cross-dataset identification pools experiments that differ in task, response scale, era and viewing conditions. The gray data lack participant IDs, so resampling is by triad. Aggregation is L1 (city-block), not Riemannian L2. The receptor compression is fixed. Sensitivity CIs and synthetic power both use test-unit intervals, which ignore training variability. *What would resolve the question:* 2AFC triads with the design of [2, 25] along cone-isolating directions and non-isoluminant L/M mixtures, where the L, M and $L\pm M$ axes disagree, including near-collinear triples off the isoluminant plane so that DR is not aliased with locus curvature.

References

[1] James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: composable transformations of Python+NumPy programs. http://github.com/jax-ml/jax, 2018.

[2] Roxana Bujack, Emily Teti, Jonah Miller, Elektra Caffrey, and Terece L. Turton. The non-Riemannian nature of perceptual color space. *Proceedings of the National Academy of Sciences*, 119(18):e2119753119, 2022. doi: 10.1073/pnas.2119753119.

[3] Roxana Bujack, Emily Teti, Jonah Miller, Elektra Caffrey, and Terece L. Turton. lanl/color: Empirical data from judgments of achromatic color differences and analysis code. https://github.com/lanl/color, 2022. BSD-3-Clause license.

[4] Roxana Bujack, Emily N. Stark, Terece L. Turton, Jonah Maxwell Miller, and David H. Rogers. The geometry of color in the light of a non-Riemannian space. *Computer Graphics Forum (Proc. EuroVis)*, 2025. doi: 10.1111/cgf.70136.

[5] Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyou Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995.

[6] CIE. Fundamental chromaticity diagram with physiological axes – part 1. Technical Report CIE 170-1:2006, Commission Internationale de l'Éclairage, 2006.

[7] CIE. Colorimetry, 4th edition. Technical Report CIE 015:2018, Commission Internationale de l'Éclairage, 2018.

[8] Colour & Vision Research Laboratory. CVRL database: CIE 2006 2-deg LMS fundamentals (linss2_10e_1) and CIE 1931 2-deg CMFs (ciexyz31_1). http://www.cvrl.org, 2006. University College London; tables of the CIE 170-1:2006 fundamentals.

[9] Jan de Leeuw and Patrick Mair. Multidimensional scaling using majorization: SMACOF in R. *Journal of Statistical Software*, 31(3):1–30, 2009.

[10] Andrew M. Derrington, John Krauskopf, and Peter Lennie. Chromatic mechanisms in lateral geniculate nucleus of macaque. *Journal of Physiology*, 357:241–265, 1984.

[11] Fritz Ebner and Mark D. Fairchild. Development and testing of a color space (IPT) with improved hue uniformity. In *Proc. IS&T/SID Sixth Color Imaging Conference*, pages 8–13, 1998.

[12] Gösta Ekman. Dimensions of color vision. *Journal of Psychology*, 38:467–474, 1954.

[13] Frederick N. Fritsch and Ralph E. Carlson. Monotone piecewise cubic interpolation. *SIAM Journal on Numerical Analysis*, 17(2):238–246, 1980.

[14] Carl E. Helm. Multidimensional ratio scaling analysis of perceived color relations. *Journal of the Optical Society of America*, 54(2):256–262, 1964.

[15] Chingis A. Izmailov and Evgeny N. Sokolov. Spherical model of color and brightness discrimination. *Psychological Science*, 2:249–260, 1991.

[16] Deane B. Judd. Interval scales, ratio scales, and additive scales for the sizes of differences perceived between members of a geodesic series of colors. *Journal of the Optical Society of America*, 57:380–386, 1967.

[17] Kenneth Knoblauch and Laurence T. Maloney. MLDS: Maximum likelihood difference scaling in R. *Journal of Statistical Software*, 25(2):1–26, 2008.

[18] John Krauskopf, David R. Williams, and David W. Heeley. Cardinal directions of color space. *Vision Research*, 22:1123–1131, 1982.

[19] David L. MacAdam. Nonlinear relations of psychometric scale values to chromaticity differences. *Journal of the Optical Society of America*, 53:754–757, 1963.

[20] Donald I. A. MacLeod and Robert M. Boynton. Chromaticity diagram showing cone excitation by stimuli of equal luminance. *Journal of the Optical Society of America*, 69:1183–1186, 1979.

[21] Laurence T. Maloney and Joong Nam Yang. Maximum likelihood difference scaling. *Journal of Vision*, 3(8):573–585, 2003.

[22] Vladimir Molchanov. Riemannian inhomogeneity and anisotropy of perceptual color space. In *EuroVis 2025 – Short Papers*. The Eurographics Association, 2025. doi: 10.2312/evs.20251080.

[23] Roger N. Shepard. Toward a universal law of generalization for psychological science. *Science*, 237:1317–1323, 1987.

[24] Vivianne C. Smith and Joel Pokorny. Spectral sensitivity of the foveal cone photopigments between 400 and 500 nm. *Vision Research*, 15:161–171, 1975.

[25] Emily Stark, Terece L. Turton, and Roxana Bujack. Diminishing returns in perceptual color space – now in color. In *EuroVis 2025 – Short Papers*. The Eurographics Association, 2025. doi: 10.2312/evs.20251081.

[26] Andrew Stockman and Lindsay T. Sharpe. The spectral sensitivities of the middle- and long-wavelength-sensitive cones derived from measurements in observers of known genotype. *Vision Research*, 40:1711–1737, 2000.

[27] Emily Stark Teti, Terece L. Turton, Jonah Maxwell Miller, and Roxana Bujack. Maximum likelihood estimation of difference scaling functions for suprathreshold judgments. *Journal of Vision*, 22(10):9, 2022. doi: 10.1167/jov.22.10.9.

[28] Louis L. Thurstone. A law of comparative judgment. *Psychological Review*, 34:273–286, 1927.

[29] Christiane B. Wiebel, Guillermo Aguilar, and Marianne Maertens. Maximum likelihood difference scales represent perceptual magnitudes and predict appearance matches. *Journal of Vision*, 17(4):1, 2017. doi: 10.1167/17.4.1. Data: `https://osf.io/58jz3/`, CC BY 4.0.

[30] Günter Wyszecki and Walter S. Stiles. *Color Science: Concepts and Methods, Quantitative Data and Formulae*. Wiley, 2nd edition, 1982.

A Cone conversion: derivation and verification

Transform. Table 2 gives the derived $\mathbf{M}$ (LMS $= \mathbf{M}\,$XYZ, CIE 1931 XYZ with white $Y = 100$, LMS in units of the peak-normalised energy fundamentals). The L and M rows are identical in the two variants. The unconstrained S row has small $\bar{x}$, $\bar{y}$ coefficients that give long-wavelength lights a spurious S excitation. At 600 nm, for a light at $Y = 31.7$, the derived S is 0.377 (unconstrained) and 0.022 (primary), against the exact value 0.0009. In the von Kries-compressed units the models see, the maximal long-wavelength ($\geq$550 nm) S error is 15.5 units (unconstrained) and 1.0 (primary). The L, M regression error peaks at 434–445 nm (10.5 units, from the CIE 1931 short-wavelength $\bar{y}$; median relative error 2.4%). This is why Ekman's monochromatic stimuli use the exact fundamentals. The CIE 2006 luminance $V = 0.6899\,\bar{l} + 0.3483\,\bar{m}$ correlates $r = 0.9988$ with $\bar{y}$.

Verification. Each white (C, D65, E) maps to von Kries $(1, 1, 1)$, compressed $(100, 100, 100)$ and CIELAB $(100, 0, 0)$, and a neutral stimulus at $L^* = 50$ maps to $(50, 50, 50)$ under every white. The sRGB primaries reproduce their reference CIELAB values to within 0.008 units, which checks our sRGB matrix and CIELAB code. The 14 spectral lights are compared with direct lookup in `logs/verification_spectral_lights.csv`. The L-cone luminance shares are $\alpha = 0.700$ (Illuminant C) and 0.708 (E, direct). Converted coordinates of every stimulus in every frame are in `logs/stimuli_cone_coordinates.csv`.

Table 2: Derived XYZ→LMS matrices and per-cone R^2 (390–780 nm).

	primary $\mathbf{M}$ (S row on $\bar{z}$ only)				unconstrained OLS (S8)		
	X	Y	Z		X	Y	Z
L	0.26827	0.84661	−0.03489	L	0.26827	0.84661	−0.03489
M	−0.38594	1.16488	0.10285	M	−0.38594	1.16488	0.10285
S	0.00000	0.00000	0.53655	S	0.02125	−0.02455	0.53416
R^2	0.9992	0.9992	0.9769	R^2	0.9992	0.9992	0.9775

A.1 Fixed bases of the M3 panel

DKL opponent: luminance $(\alpha, 1-\alpha, 0)$, $L{-}M$ $(1, -1, 0)$ and $S{-}$luminance $(-\alpha, -(1-\alpha), 1)$, in von Kries units, where α is the L-cone share of the white's CIE 2006 luminance $V = 0.6899\,\bar{l} + 0.3483\,\bar{m}$. The chromatic axes are silent for neutral stimuli and the luminance axis is silent at

isoluminance. *Equal-weight opponent*: $((L{+}M)/2,\ L{-}M,\ S{-}(L{+}M)/2)$. ***sRGB*** and ***CIE RGB***: rows of $\mathbf{M}_{\mathrm{RGB}\leftarrow\mathrm{XYZ}}\mathbf{M}^{-1}\mathrm{diag}(\mathrm{LMS}_w)$ with the space's own white (D65; E), i.e. the linear-light channels of the adapted signal. ***CIELAB*** and ***CIE LCh*** use their own coordinate differences with the standard transform and dataset white [7], with $\Delta H^* = 2\sqrt{C_1 C_2}\sin(\Delta h/2)$. Each axis of each basis is divided by its fixed design RMS: the root-mean-square projected difference over each dataset's pairs, averaged with equal dataset weight. Projections that are zero up to rounding are set to exactly 0.

Table 3: Fixed linear bases acting on (compressed) von Kries cone differences. sRGB and CIE RGB rows are the linear-light channels of the adapted signal re-illuminated by the space's own white; they do not depend on the dataset white. Each axis is then divided by its fixed design RMS.

basis (white)	axis	u_L	u_M	u_S
opponent (DKL) (C)	Lum	0.6998	0.3002	0.0000
	L−M	1.0000	−1.0000	0.0000
	S−Lum	−0.6998	−0.3002	1.0000
opponent (DKL) (D65)	Lum	0.6983	0.3017	0.0000
	L−M	1.0000	−1.0000	0.0000
	S−Lum	−0.6983	−0.3017	1.0000
opponent (equal-weight)	(L+M)/2	0.5000	0.5000	0.0000
	L−M	1.0000	−1.0000	0.0000
	S−(L+M)/2	−0.5000	−0.5000	1.0000
sRGB	R	5.2942	−4.4934	0.1992
	G	−0.6739	1.8847	−0.2108
	B	−0.0232	−0.1450	1.1681
CIE RGB	R	4.0691	−3.0933	0.0242
	G	−0.0838	1.1295	−0.0457
	B	0.0008	−0.0114	1.0106

B Optimisation diagnostics

Table 4 summarises the 41,410 fits of the primary specification. Every L-BFGS-B run terminated with a success flag. For every stored optimum we recomputed the projected gradient with training weights rebuilt from the logged seeds: at most 1.1% of the fits of any model exceed 0.1, and the largest gradient relative to the objective is 7.1×10^{-4}. The share of fits whose best optimum was reached by at least two starts is lowest for M2 (67%). This reflects the flat, multimodal curvature directions documented in §4, not failed convergence. An earlier version of the gray-axis replication did suffer an optimisation failure: one free-scale MLDS fit converged to a degenerate non-monotone scale. It was fixed by adding a deterministic start to every fit, and the failure and fix are recorded in `attempts_log.md`.

Table 4: Convergence diagnostics, primary specification (10 folds × 101 fits per model).

model	fits	success	best by ≥2 starts	median proj. grad.	frac. proj. grad. >0.1
M0	1010	1.000	1.000	1.1e−04	0.000
M1	1010	1.000	1.000	1.6e−04	0.000
G1	1010	1.000	0.965	2.0e−04	0.000
C2	1010	1.000	0.908	2.7e−04	0.000
M2	1010	1.000	0.670	3.6e−04	0.003
M3_opp	1010	1.000	0.782	4.4e−04	0.011
M3_oppEW	1010	1.000	0.886	4.1e−04	0.002
M3_srgb	1010	1.000	0.971	3.0e−04	0.000
M3_ciergb	1010	1.000	0.782	3.5e−04	0.003
M3_lab	1010	1.000	0.965	4.2e−04	0.005
M3_hcl	1010	1.000	0.702	5.8e−04	0.009

C All models, primary specification

Table 5: Held-out log-likelihood of every fitted model (sum over the 10 test folds) and its per-dataset decomposition. The last column gives the difference to M1 with its paired double-bootstrap 95% CI. $M0_B$ and $M1_B$ are the additive and isotropic-DR models in basis B; 2ch_B_ij_k ties the curvatures of axes i, j of basis B.

model	total ℓ	gray 2AFC	MLDS	Helm	Ekman	Δ vs M1 [95% CI]
M0	−65447.6	−60321.4	−4833.6	−344.6	52.0	−521.9 [−728.8, −322.0]
M1	−64925.6	−59905.9	−4755.7	−309.2	45.2	(reference)
G1	−64930.7	−59906.9	−4756.1	−312.9	45.3	−5.0 [−10.1, 4.6]
C2	−64967.8	−59925.1	−4755.6	−334.9	47.7	−42.1 [−132.7, 26.6]
M2	−64936.8	−59976.2	−4754.5	−240.3	34.3	−11.1 [−184.6, 102.8]
M3_opp	−64941.2	−59897.5	−4755.1	−296.3	7.7	−15.6 [−265.5, 291.9]
M3_oppEW	−64932.5	−59896.5	−4755.0	−298.7	17.7	−6.9 [−252.4, 296.5]
M3_srgb	−64908.4	−59950.7	−4755.0	−240.6	37.9	+17.2 [−237.8, 262.3]
M3_ciergb	−64881.8	−59948.6	−4754.3	−222.8	43.9	+43.9 [−182.9, 279.0]
M3_lab	−64904.0	−59897.1	−4755.2	−262.5	10.9	+21.7 [−212.5, 305.8]
M3_hcl	−64844.3	−59896.3	−4755.0	−221.7	28.7	+81.3 [−414.8, 294.4]
M0_opp	−65524.1	−60321.4	−4833.6	−419.0	49.8	−598.5 [−870.1, −278.7]
M0_oppEW	−65510.8	−60321.4	−4833.6	−413.3	57.5	−585.2 [−817.4, −288.6]
M0_srgb	−65464.5	−60321.4	−4833.6	−367.3	57.8	−538.8 [−815.6, −242.9]
M0_ciergb	−65454.8	−60321.4	−4833.6	−359.4	59.6	−529.2 [−760.2, −251.0]
M0_lab	−65503.8	−60321.4	−4833.6	−362.6	13.8	−578.1 [−852.3, −263.7]
M0_hcl	−65510.1	−60321.4	−4833.6	−399.9	44.8	−584.5 [−978.2, −244.9]
M1_opp	−64940.8	−59903.0	−4755.1	−302.2	19.6	−15.1 [−209.1, 260.9]
M1_oppEW	−64901.2	−59903.0	−4755.2	−259.6	16.6	+24.4 [−152.4, 317.0]
M1_srgb	−64919.9	−59907.2	−4756.1	−294.1	37.5	+5.7 [−291.0, 243.9]
M1_ciergb	−64943.6	−59906.0	−4755.9	−327.1	45.4	−17.9 [−306.2, 205.4]
M1_lab	−64872.8	−59903.0	−4755.2	−231.0	16.3	+52.8 [−151.0, 326.3]
M1_hcl	−64840.7	−59903.1	−4755.2	−211.3	28.9	+84.9 [−157.8, 335.4]
2ch_ciergb_01_2	−64946.1	−59906.5	−4756.1	−328.9	45.4	−20.5 [−319.4, 195.0]
2ch_ciergb_02_1	−64937.5	−59906.9	−4756.2	−319.5	45.1	−11.8 [−310.8, 197.1]
2ch_ciergb_12_0	−64880.4	−59948.2	−4754.3	−222.0	44.2	+45.2 [−193.0, 279.6]
2ch_hcl_01_2	−64844.3	−59896.3	−4755.0	−221.7	28.7	+81.3 [−248.2, 301.8]
2ch_hcl_02_1	−64921.7	−59900.1	−4755.2	−294.6	28.2	+3.9 [−463.7, 274.8]
2ch_hcl_12_0	−64844.3	−59896.3	−4755.0	−221.7	28.7	+81.3 [−246.0, 302.9]
2ch_lab_01_2	−64870.8	−59896.4	−4755.0	−235.3	15.9	+54.9 [−175.9, 319.9]
2ch_lab_02_1	−64901.3	−59896.6	−4755.0	−260.9	11.2	+24.4 [−199.5, 313.1]
2ch_lab_12_0	−64873.2	−59896.3	−4755.0	−238.2	16.3	+52.5 [−188.3, 311.1]
2ch_oppEW_01_2	−64925.0	−59897.1	−4755.0	−290.9	18.0	+0.6 [−227.7, 287.9]
2ch_oppEW_02_1	−64928.3	−59896.6	−4755.0	−295.3	18.5	−2.7 [−219.5, 309.1]
2ch_oppEW_12_0	−64936.6	−59896.1	−4755.0	−301.1	15.7	−10.9 [−260.2, 271.5]
2ch_opp_01_2	−64940.6	−59896.6	−4755.1	−306.9	17.9	−15.0 [−235.0, 266.7]
2ch_opp_02_1	−64940.1	−59899.0	−4755.6	−294.2	8.7	−14.5 [−245.5, 287.7]
2ch_opp_12_0	−64905.3	−59896.3	−4755.0	−270.6	16.6	+20.3 [−227.4, 289.7]
2ch_srgb_01_2	−64934.7	−59911.7	−4757.1	−300.7	34.8	−9.1 [−303.2, 227.0]
2ch_srgb_02_1	−64930.9	−59909.6	−4757.1	−301.6	37.4	−5.2 [−309.7, 227.7]
2ch_srgb_12_0	−64909.2	−59951.4	−4755.1	−240.7	38.0	+16.4 [−238.7, 265.1]

D Sensitivity specifications

All pre-registered secondary specifications were run on the same folds and are reported in full in Table 6. No specification was selected among them. S1–S8 use full-training fits, so their CIs come from a test-unit bootstrap that ignores training variability; on P such intervals are 0.59 times the width of the double-bootstrap intervals, so S1–S8 intervals are optimistic. S8 is the conversion exactly as pre-registered (unconstrained M, Ekman through M); S2 refits every model separately on each dataset and sums the per-dataset held-out log-likelihoods, which removes every cross-dataset tie.

E Curvature control: arc vs chord per model

F Synthetic identifiability study: details

All data in this section are **synthetic**. Each generator's responses were simulated at the real stimulus configurations, trial counts and observation models: binomial draws for the two triad datasets, and Gaussian draws on the log-distance and similarity scales for Helm and Ekman. Nuisance parameters were taken from the corresponding real-data fit. Each synthetic data set was analysed with the same 10 folds and models as the real data, using full-training fits (4 random starts plus the deterministic start) and test-unit bootstrap CIs. Table 8 gives the percentage of the synthetic data sets in which each decision rule fired. Rates in a column whose generator does not contain the effect are false-positive rates.

Table 6: **All specifications** (held-out $\Delta\ell$, 95% CI; * CI excludes 0). P uses the double bootstrap; S1–S8 use test-unit bootstrap CIs. The cone-alignment column is selection-aware. Specifications: P: primary (shared latent, vK + compression, log families); S1: outer a per dataset; S2: fully per-dataset fits (sum); S3: no receptor compression (Helm+Ekman only); S4: luminance-weighted cone scaling (also alters the dark-gray scale via the CIE toe); S5: power families (inner u^p, outer y^γ); S6: Ekman white D65; S7: lapse rate in binary models; S8: unconstrained $\mathbf{M}$, Ekman via XYZ (as pre-registered).

spec	M1−M0 (DR)	M2−M1 (aniso.)	M2−best M3 (sel.-aware)	best control	C2−M1	C2−best 2-ch
P	+521.9*	−11.1	−92.5	CIE LCh	−42.1	−123.5
	[322.0, 728.8]	[−184.6, 102.8]	[−406.5, 61.8]		[−132.7, 26.6]	[−423.4, 3.5]
S1	+649.4*	+4.0	+10.9	CIE RGB	+4.0	+3.9
	[416.6, 888.4]	[−3.5, 11.8]	[−137.5, 92.1]		[−3.5, 11.8]	[−138.0, 69.9]
S2	+658.2*	+33.3	−73.6*	sRGB	+41.9	−73.5*
	[427.0, 894.8]	[−20.5, 91.8]	[−178.6, −30.0]		[−0.2, 87.9]	[−170.4, −22.0]
S3	+148.8*	+10.4	−72.7*	CIE LCh	−12.7	−157.2*
	[17.1, 284.4]	[−40.4, 58.9]	[−214.1, −21.9]		[−44.0, 16.5]	[−256.9, −60.8]
S4	+578.6*	+150.9*	−547.8*	opp. (equal-wt)	−20.5	−719.3*
	[377.1, 780.8]	[39.6, 282.4]	[−887.8, −188.6]		[−81.9, 41.8]	[−1037.6, −393.7]
S5	+508.2*	−5.6*	−96.8*	CIE LCh	+0.0	−93.3*
	[334.7, 694.8]	[−11.6, −1.3]	[−208.4, −36.4]		[0.0, 0.0]	[−201.1, −30.7]
S6	+522.0*	−11.2	−92.8	CIE LCh	−42.2	−123.8*
	[332.4, 715.7]	[−117.9, 98.8]	[−266.6, 65.5]		[−87.0, 3.2]	[−253.8, −49.4]
S7	+522.3*	−41.2	−108.3*	CIE RGB	−14.6*	−93.4*
	[323.1, 721.9]	[−131.2, 38.6]	[−247.4, −33.1]		[−25.1, −3.2]	[−219.9, −12.3]
S8	+524.6*	−6.9	−91.1	CIE LCh	−15.1*	−99.4*
	[333.2, 718.4]	[−114.9, 105.1]	[−263.3, 68.2]		[−26.2, −4.3]	[−221.1, −19.4]

Table 7: Held-out log-likelihood of the hue-locus datasets with arc-length (before control) and chord (after control) inputs. Positive chord−arc means the additive-metric chord geometry predicts better.

	Helm held-out ℓ			Ekman held-out ℓ		
model	arc (before)	chord (after)	chord−arc	arc	chord	chord−arc
M0	−374.0	−344.6	+29.4	56.8	52.0	−4.8
M1	−310.0	−309.2	+0.8	44.7	45.2	+0.5
G1	−310.0	−312.9	−2.8	44.7	45.3	+0.6
C2	−157.9	−334.9	−177.0	53.9	47.7	−6.2
M2	−178.5	−240.3	−61.8	51.6	34.3	−17.3
M3_opp	−184.3	−296.3	−112.0	23.1	7.7	−15.4
M3_oppEW	−183.6	−298.7	−115.1	21.9	17.7	−4.2
M3_srgb	−246.2	−240.6	+5.6	40.3	37.9	−2.4
M3_ciergb	−216.5	−222.8	−6.3	45.3	43.9	−1.4
M3_lab	−240.3	−262.5	−22.2	12.4	10.9	−1.5
M3_hcl	−273.0	−221.7	+51.3	31.3	28.7	−2.6

G Pre-registration and deviations

The model family, gauge, observation models, folds, bootstrap and selection-aware protocol, identifiability plan and confound plan were fixed in the time-stamped `analysis_plan.md` (02:57 UTC) before any fitting code existed. Every run, failure and change is logged with a timestamp in `attempts_log.md`. The deviations are: (i) *cone conversion*. The pre-registered spectral verification showed spurious long-wavelength S excitation from the unconstrained matrix, so the primary transform constrains the S row to $\bar{z}$ and Ekman's monochromatic lights use the exact fundamentals. The literal pre-registered pipeline is reported as S8 and gives the same verdicts (Table 6). (ii) *Optimisation safeguard.* After a degenerate MLDS optimum in the first replication run, every fit also starts from a deterministic, well-posed point. The replication was rerun, the primary run was restarted at 11% completion before any of its output was analysed, and all reported numbers come from the safeguarded runs. (iii) *Near-geodesic segments.* The rule "runs of $\geq$4 stimuli with interior angles $\geq 170°$" was applied to all qualifying runs rather than only the longest, with the threshold unchanged. (iv) *Generators.* The best two-channel control coincides with the best M3 member at its MLE, so it was not simulated separately. (v) *Bugs.* A power-family offset error (S5 only) and a silent-axis scaling error in single-dataset fits (S2 only) were found while checking implausible sensitivity results. Both were fixed and the affected specifications rerun; all other specifications were verified to be unchanged (log-likelihood changes $\leq 10^{-9}$). Discarded runs are kept in `results/*_buggy`. (vi) *Not*

Table 8: **Decision rates (%) in synthetic data** (100 data sets per generator). A decision fires when the 95% CI excludes 0 in the stated direction.

generator (**synthetic**)	M2>M1	C2>M1	M2>C2	M2>DKL	M2<DKL	M2>best M3	M2<best M3	C2>best 2-ch
M2 (MLE)	90	11	83	100	0	61	0	1
M2 designed distinct	50	50	12	98	0	66	0	48
M1 (isotropic)	0	0	6	90	0	53	0	45
M3 DKL opponent	100	97	32	0	100	0	100	0
M3 CIE LCh	100	100	0	0	100	0	100	0
C2 (MLE)	2	2	2	83	0	30	0	19

pre-registered and not used: one exploratory design check (canonical correlations for a hypothetical augmented design) was judged uninformative and is reported only in the log.

H Gray-axis replication: details

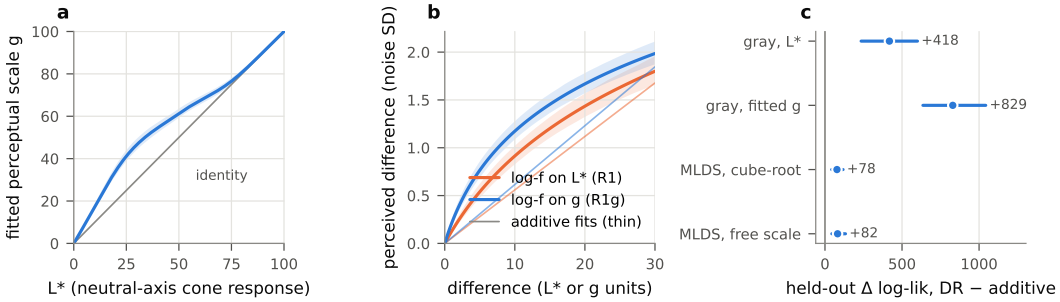


Figure 5: **Gray-axis anchor.** (a) Fitted perceptual scale $g(L^*)$ (R1g, 95% band over 1000 training-fold bootstrap fits), steepest near the dark-blue background as in 2022 (crispening). (b) Fitted difference scaling (thick) against the additive fits (thin), in noise-SD units. (c) Held-out log-likelihood gain of log-f over additive models (paired double-bootstrap 95% CI), for both gray datasets.

In the triad-disjoint CV, the 2022 accuracy metric is 0.897, 0.904, 0.921 and 0.943 for R0, R1, R0g and R1g. In the within-triad splits R1 beats R0 in 10/10 splits and R1g beats R0g in 10/10. The free-scale MLDS contrast W0g−W0 is +46.9 [−26.6, 89.1]. The fitted g passes through 41.2 at L^*=25, so it rises most steeply near the dark-blue background ($L^* \approx 20$), matching the crispening reading of 2022. The joint fit of g and f gives a larger log shape than 2022's sequential procedure; the 2022-normalised amplitude b matches.